

KERALA DIPLOMA CURRICULUM REVISION 2026

Course 2421

II

As specified in the official PDF

Source-grounded

Food Processing Technology

Study the official course objectives in the source syllabus.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 2421 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 2421.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma Programme
Course code	2421
Course title	Food Processing Technology
Semester	II
Course category	As specified in the official PDF
Periods per week	As specified in the official PDF
Periods per semester	As specified in the official PDF
Credits	As specified in the official PDF
Prerequisite	Topic Course code Course Title Semester Basic knowledge of different components of food 1421 Fundamentals of food processing technology I Course Outcome CO Course Outcome Blooms Taxonomy Level CO1 Explain the concept of raw material preparation by high temperature processing Understand CO2 Discuss size reduction principles and equipment for solid foods Understand CO3 Illustrate the importance of drying, refrigeration and freezing during processing Understand CO4 Summarize the concepts of extrusion and minimal processing techniques in food technology sector. Understand CO-PO mapping COURSE ARTICULATION MATRIX PROGRAM OUTCOMES
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES**Objectives, outcomes and mapping****Course objectives**

- Study the official course objectives in the source syllabus.

Course outcomes

CO	Official outcome	Study level
CO1	3 - - - - -	See official syllabus
CO2	3 - - - - -	See official syllabus
CO3	3 - - - - -	See official syllabus
CO4	3 - - - - - Total 3 - - - - - Correlation Level 3 - - - - - Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - "-" Teaching and Assessment Scheme Examination Scheme Teaching Scheme/Week Self-learning Total Credits Theory Practical (SS)/Week Hours/Semester C Total L T P S CIE ESE Total CIE ESE Total 4 0 0 0 6 60 4 40 60 100 0 0 0 100 L: Lecture (One unit is of one-hour duration with one credit), T: Tutorial (One unit is of one-hour duration with one credit), P: Practical (One unit is of one-hour duration with 0.5 credit), S: Project (One unit is of one-hour duration with 0.5 credit) SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination Syllabus - Major Topics Diploma Curriculum Revision 2026 2421 Page #1 Instructional Module Title Major Topics Hours L T Raw material preparation and thermal	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3 - - - - -

Row	Extracted official mapping
CO2	CO2 3 - - - - -
CO3	CO3 3 - - - - -
CO4	CO4 3 - - - - -

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	cleaning techniques,pasteurization&blanching 13 0	3	As specified
Module 2	13 0	3	As specified
Module 3	minimal processing techniques in food 17 0	4	As specified
Module 4	drying,refrigeration ,chilling& freezing 17 0	6	As specified

MODULE 1**cleaning****techniques,pasteurization&blanching 13 0**

This module is presented from the official syllabus scope for Food Processing Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: cleaning techniques,pasteurization&blanching 13 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ cleaning techniques,pasteurization&blanching 13 0

cleaning techniques,pasteurization&blanching 13 0 is an official topic in Module 1 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify cleaning techniques,pasteurization&blanching 13 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, cleaning techniques,pasteurization&blanching 13 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What cleaning techniques,pasteurization&blanching 13 0 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-പാലി ക്ലീനിംഗ്
 techniques, pasteurization & blanching 13 0 പാലി ക്ലീനിംഗ്
 definition/meaning പാലി ക്ലീനിംഗ്. പാലി ക്ലീനിംഗ് പാലി ക്ലീനിംഗ്, steps,
 application പാലി ക്ലീനിംഗ് പാലി ക്ലീനിംഗ്. Technical terms English-പാലി ക്ലീനിംഗ്
 പാലി ക്ലീനിംഗ്.

▼ processing in food industries

processing in food industries is an official topic in Module 1 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify processing in food industries using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, processing in food industries should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What processing in food industries means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓഡിറ്റിംഗ് പ്രക്രിയയിൽ ഭക്ഷണ സംസ്കരണ വ്യവസ്ഥകളിൽ

പ്രധാനപ്പെട്ടവയെക്കുറിച്ച് നിങ്ങളുടെ definition/meaning എഴുതേണ്ടതാണ്. നിങ്ങളുടെ നിങ്ങളുടെ നിങ്ങളുടെ, steps, application എഴുതേണ്ടതാണ് നിങ്ങളുടെ നിങ്ങളുടെ. Technical terms English-ൽ നിങ്ങളുടെ നിങ്ങളുടെ എഴുതേണ്ടതാണ്.

▼ Size reduction and homogenization in food principles of size reduction, equipment used for size

Size reduction and homogenization in food principles of size reduction, equipment used for size is an official topic in Module 1 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Size reduction and homogenization in food principles of size reduction, equipment used for size using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, size reduction and homogenization in food principles of size reduction, equipment used for size should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What Size reduction and homogenization in food principles of size reduction, equipment used for size means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Size reduction and homogenization in food principles of size reduction, equipment used for size definition/meaning. Steps, application. Technical terms English-Malayalam.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2**13 0**

This module is presented from the official syllabus scope for Food Processing Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: 13 0
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

▼ processing reduction,homogenization in food processing

processing reduction,homogenization in food processing is an official topic in Module 2 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify processing reduction,homogenization in food processing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, processing reduction,homogenization in food processing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What processing reduction,homogenization in food processing means in the official course context

▼ Summarize the concepts of extrusion and

Summarize the concepts of extrusion and is an official topic in Module 2 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Summarize the concepts of extrusion and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, summarize the concepts of extrusion and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Summarize the concepts of extrusion and means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-
Summarize the concepts of extrusion and definition/meaning.
 , steps, application . Technical terms English-
 .

▼ extrusion process,minimal processing of fruits and vegetables

extrusion process,minimal processing of fruits and vegetables is an official topic in Module 2 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify extrusion process,minimal processing of fruits and vegetables using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, extrusion process,minimal processing of fruits and vegetables should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What extrusion process,minimal processing of fruits and vegetables means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എക്സ്ട്രൂഷൻ പ്രക്രിയ,മിനималь പ്രോസസ്സിംഗ് ഫ്രൂട്ട്സ് ആൻഡ് വെജിറ്റബിൾസ് എന്നതിന്റെ definition/meaning എഴുതുക. എന്നതിന്റെ components, steps, application എഴുതുക. Technical terms English-ൽ എഴുതുക.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3**minimal processing techniques in food 17 0**

This module is presented from the official syllabus scope for Food Processing Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: minimal processing techniques in food 17 0
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

▼ minimal processing techniques in food 17 0

minimal processing techniques in food 17 0 is an official topic in Module 3 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify minimal processing techniques in food 17 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, minimal processing techniques in food 17 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What minimal processing techniques in food 17 0 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- minimal processing techniques in food
 17 0 definition/meaning .
 , steps, application . Technical
 terms English- .

▼ ,hurdle technology

,hurdle technology is an official topic in Module 3 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify ,hurdle technology using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, ,hurdle technology should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What ,hurdle technology means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പുറം, hurdle technology പരീക്ഷിക്കുന്നതിനായി നിർവ്വചനം/അർത്ഥം പരീക്ഷിക്കുന്നതിനായി. പരീക്ഷിക്കുന്നതിനായി പരീക്ഷിക്കുന്നതിനായി, steps, application പരീക്ഷിക്കുന്നതിനായി പരീക്ഷിക്കുന്നതിനായി. Technical terms English-ൽ പരീക്ഷിക്കുന്നതിനായി പരീക്ഷിക്കുന്നതിനായി.

▼ technology sector.

technology sector. is an official topic in Module 3 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify technology sector. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, technology sector. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What technology sector. means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓഡിയോ technology sector. ഓഡിയോ ഓഡിയോ definition/meaning ഓഡിയോ. ഓഡിയോ ഓഡിയോ ഓഡിയോ, steps, application ഓഡിയോ ഓഡിയോ. Technical terms English-ഓ ഓഡിയോ ഓഡിയോ.

▼ Illustrate the importance of drying,

Illustrate the importance of drying, is an official topic in Module 3 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the importance of drying, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the importance of drying, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the importance of drying, means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓരോന്നും ഉപയോഗിക്കുക. Illustrate the importance of drying, ഓരോന്നും ഉപയോഗിക്കുക definition/meaning ഓരോന്നും ഉപയോഗിക്കുക. ഓരോന്നും ഉപയോഗിക്കുക, steps, application ഓരോന്നും ഉപയോഗിക്കുക. Technical terms English-ഓരോന്നും ഉപയോഗിക്കുക.

► **Official syllabus detail and terminology (expand in digital version)**

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4**drying,refrigeration ,chilling& freezing 17 0**

This module is presented from the official syllabus scope for Food Processing Technology. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: drying,refrigeration ,chilling& freezing 17 0
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

▼ drying,refrigeration ,chilling& freezing 17 0

drying,refrigeration ,chilling& freezing 17 0 is an official topic in Module 4 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify drying,refrigeration ,chilling& freezing 17 0 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, drying,refrigeration ,chilling& freezing 17 0 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What drying,refrigeration ,chilling& freezing 17 0 means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓരോന്നും drying, refrigeration, chilling & freezing 17 0 ഓരോന്നും definition/meaning ഓരോന്നും. ഓരോന്നും ഓരോന്നും ഓരോന്നും, steps, application ഓരോന്നും ഓരോന്നും. Technical terms English-ഓരോന്നും ഓരോന്നും.

▼ refrigeration and freezing during processing

refrigeration and freezing during processing is an official topic in Module 4 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify refrigeration and freezing during processing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, refrigeration and freezing during processing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What refrigeration and freezing during processing means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-റഫ്രിജറേഷൻ and freezing during processing ഫുഡ് പ്രോസസ്സിംഗ് റഫ്രിജറേഷൻ definition/meaning റഫ്രിജറേഷൻ. റഫ്രിജറേഷൻ ഫുഡ് പ്രോസസ്സിംഗ്, steps, application ഫുഡ് പ്രോസസ്സിംഗ് റഫ്രിജറേഷൻ. Technical terms English-റ ഫുഡ് പ്രോസസ്സിംഗ്.

▼ Detailed Syllabus

Detailed Syllabus is an official topic in Module 4 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Detailed Syllabus using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, detailed syllabus should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Detailed Syllabus means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
Detailed Syllabus
definition/meaning
application
steps,
Technical terms English-
.

▼ Mode of

Mode of is an official topic in Module 4 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mode of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mode of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mode of means in the official course context
Study lens	Principle → components or stages → result → application

Aspect	What to prepare
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-എന്നിവിടെ Mode of ഏകീകരണം എന്നതിന് definition/meaning എഴുതണം. ഏകീകരണം എങ്ങനെ നടത്തണം, steps, application എങ്ങനെ നടത്തണം എന്നിവ എഴുതണം. Technical terms English-ൽ എഴുതണം.

▼ Mapped Taxonomy Instructional

Mapped Taxonomy Instructional is an official topic in Module 4 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Mapped Taxonomy Instructional using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, mapped taxonomy instructional should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Mapped Taxonomy Instructional means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-
 definition/meaning .
 , steps, application . Technical terms
 English- .

▼ Subtopic Student Learning Outcome delivery

Subtopic Student Learning Outcome delivery is an official topic in Module 4 of Food Processing Technology. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Subtopic Student Learning Outcome delivery using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, subtopic student learning outcome delivery should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Subtopic Student Learning Outcome delivery means in the official course context

Aspect	What to prepare
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-Subtopic Student Learning Outcome delivery ഫോറം definition/meaning ഫോറം. ഫോറം ഫോറം, steps, application ഫോറം ഫോറം. Technical terms English-ഫോറം ഫോറം.

► Official syllabus detail and terminology (expand in digital version)

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE**Concept and formula bank****Answer structure**

Definition → Principle →
Components/steps →
Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure
→ Observation → Result →
Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION**Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING**Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

- for Self-Learning
- Week Self-Learning description Hours Module I
- Think-Pair-Share Activity Ask students: “Why is blanching done before freezing vegetables?”Think individually (2 minutes)
- Discuss with a partner (3 minutes),Share answers with class
- Process Sequencing Activity - Provide jumbled steps of pasteurization / sterilization.Arrange steps in correct order and
- explain each step. Module II
- Equipment-Principle Matching Provide two columns:
- Column A: Hammer mill, Roller mill, Cutter, Ball mill
- Column B: Impact, Compression, Shear, Attrition
- Match equipment with working principle and justify choice
- Object-Based Analogy Activity - Show pictures / real items (nuts, grains, flour). Ask students which equipment is suitable and Module III Comparative Group Chart
- Divide class into 3 groups: Group 1: Drying Group 2: Refrigeration Group 3: Freezing
- Each group prepares a chart with: Temperature range Shelf life Advantages Examples
- Real-Life Product Identification - Ask students to list food products they use daily that are:Dried,Refrigerated,Frozen Explain
- why each method is used. Module IV
- Video-Based Discussion - Show a short video of an extruder or MAP packaging. 12
- Role-Play Activity - Students act as:Foodtechnologist,Consumer,Manufacturer.
- Choosing between minimally processed salad and conventional processed food.
- Total Self-Learning hours 90 Learning Resources TEXT BOOK
- Title Author Publication

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

– Theory					
		CA1		CA2	
CA3	CA4	CA5	CA6		
ESE					
Mode	Attendance	Self-learning	Self-learning		
Self-learning	Self-learning	Series	Series		
Written					
activities	activities	activities	activities		
Exam		Exam	Exam		
(Module 3)	(Module 4)	(Module 1)	(Module 2)		
(theory)		(2 modules)	(2 modules)		
Duration					
2 hours	2 hours	2.5 hours			
Exam mark					
15	15	50	15	50	60
Diploma Curriculum Revision 2026					
2421					
Page #3					
Converted to					
20	20				
Marks		5			

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

► Define or explain cleaning techniques, pasteurization & blanching 13 0. (3 marks)

► Define or explain processing in food industries. (3 marks)

► Define or explain Size reduction and homogenization in food principles of size reduction, equipment used for size. (3 marks)

Module 2

► Define or explain processing reduction, homogenization in food processing. (3 marks)

► Define or explain Summarize the concepts of extrusion and. (3 marks)

► Define or explain extrusion process, minimal processing of fruits and vegetables. (3 marks)

Module 3

► Define or explain minimal processing techniques in food 17 0. (3 marks)

► Define or explain , hurdle technology. (3 marks)

► **Define or explain technology sector.. (3 marks)**

► **Define or explain Illustrate the importance of drying,.. (3 marks)**

Module 4

► **Define or explain drying,refrigeration ,chilling& freezing 17 0. (3 marks)**

► **Define or explain refrigeration and freezing during processing. (3 marks)**

► **Define or explain Detailed Syllabus. (3 marks)**

► **Define or explain Mode of. (3 marks)**

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **cleaning techniques, pasteurization & blanching** **13 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **processing reduction, homogenization in food processing.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **minimal processing techniques in food** **17 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

Module 4

1-mark definition: State the meaning of **drying, refrigeration, chilling & freezing** **17 0.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- URL Modules Covered
- https://www.ajce.in/fe/downloads/FoodTechnologysyllabus%28s5%2Cs6%29.pdf?utm_source=chatgpt.com I & II
- https://agriculture.institute/food-fundamentals-cpo/unit-operations-food-processing-techniques-applications/?utm_source=chatgpt.com III & IV

IN-PAGE SEARCH**Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 2421. Verify institutional updates against the current official curriculum.